

Do prediction intervals for satellite-based poverty estimates survive a national border? A pre-registered evaluation in West Africa

Abdoulie Balisa

Department of Statistics and Actuarial Science
Kwame Nkrumah University of Science and Technology
`abalisa@st.knust.edu.gh`

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Abstract

Satellite-based models of household wealth are trained and validated on survey clusters from one set of countries and then applied to countries with no survey at all. Their reported accuracy is therefore measured under conditions that do not match how they are used. We asked whether the uncertainty such models report remains valid across a national border, using cluster-level wealth index values from twelve West African Demographic and Health Surveys (6,706 clusters, 155,796 households) and open geospatial covariates. The Gambia was withheld from every stage of model development and evaluated once, against a threshold registered in advance.

Four hypotheses were stated before analysis and none was supported. Point accuracy degraded by about three per cent across a border rather than the predicted third ($R^2 = 0.746$ in-country under spatial blocking, 0.723 under leave-one-country-out, 0.673 on The Gambia). Mean conformal coverage also transferred: a nominal 90% interval covered 88.7% of held-out foreign clusters and 89.3% of Gambian clusters, where the pre-registered prediction was substantially less than 90%. Correlation alignment degraded point accuracy rather than improving it. The pre-registered target-country test of a coverage-reliability hypothesis returned a null at 53% power.

Two observations survive as leads rather than results. Coverage varied across held-out countries by 3.7 times what sampling noise permits (range 0.821 to 0.979 at nominal 90%), so the average conceals substantial per-country variation. On The Gambia, constant-width split conformal undercovered at nominal 50% and 80% by more than four standard errors while remaining at nominal at 90% and 95%, and adaptive conformalized quantile regression did not. Both are post-hoc and are reported as such. Coverage holding here is a contrast rather than a foregone conclusion: concurrent work measuring conformal coverage on an African satellite product under spatial shift reports it falling to 65.3% against a 90% target. We conclude that cross-border transfer of wealth models is better behaved than caution about distribution shift implies, in a region of neighbouring countries sharing a survey instrument, and that the residual risk lies in interval shape and in per-country variability rather than in average coverage.

1 Introduction

Household surveys are expensive and infrequent. The Gambia has two Demographic and Health Surveys with georeferenced clusters, in 2013 and 2019–20. Many countries have fewer. The standard response is to train a model that predicts survey-measured wealth from satellite imagery and other open geospatial data in countries where surveys exist, then apply it where they do not (Jean et al., 2016; Yeh et al., 2020; Chi et al., 2022).

The published literature reports how accurate such models are. It reports this almost always by holding out clusters from the same countries used for training. The application of interest is different: a country with no survey at all. If accuracy is estimated in-domain and the model is applied out-of-domain, reported performance may not describe the deployment setting.

Point accuracy under this shift is reasonably well studied. Yeh et al. (2020) report $R^2 = 0.67$ for asset wealth in held-out countries. Uncertainty is treated less consistently. Interval estimates are sometimes published: Chi et al. (2022) release a confidence interval with every wealth microestimate, and Kakooei and Daoud (2024) study predictive uncertainty for poverty estimation explicitly under cross-country transfer. Empirical calibration is reported rarely and narrowly. Chi et al. (2022) construct intervals from a regression on absolute residuals, note its limited explanatory power, and do not report the coverage those intervals achieve. In concurrent work, Olanrewaju and Adebayo (2026) do report coverage, at 88.4% against a 90% target, for estimates fitted on each country’s own survey. That is one figure at one nominal level for the in-country regime, and it is the closest published check we found.

A model that predicts a cluster’s wealth index and attaches a 90% prediction interval makes a falsifiable claim: across many such clusters, 90% should contain the truth. We measure whether that claim holds, and whether it survives a national border.

This paper measures it. The contribution is the measurement and the design that makes it interpretable, not the direction of the answer, which was not what we predicted.

1.1 What we predicted, and what happened

Four hypotheses were fixed before data access was granted, in a protocol committed to version control (Section 4). All four were disconfirmed. We report this rather than reframing it, and Section 5 gives each result at the strength the evidence supports.

2 Related work

2.1 Predicting wealth from satellite and digital trace data

The modern form of this problem was set by Jean et al. (2016), who used transfer learning from nighttime lights to extract features from daytime satellite imagery and predicted cluster-level consumption and assets in five African countries, reporting R^2 up to 0.56. Yeh et al. (2020) scaled the approach to roughly 20,000 villages across 23 African countries with a multispectral convolutional network, reporting $R^2 = 0.67$ for asset wealth in held-out countries, which remains the strongest published evidence that point accuracy survives a border.

Cross-country prediction has been examined directly. Lee and Braithwaite (2022) build village-level poverty maps across 25 sub-Saharan African countries and evaluate prediction for one country from others. Corral et al. (2025) assess how these models behave where no survey exists and find the errors are not random. The direction of the error tracks the characteristics of the place: living standards come out too high where the area is poor, rural or dependent on one sector, and too low where it is richer and more urban. Adding spatial structure to the model does not remove the pattern. Their finding concerns the location of the point estimate. Ours concerns the width of the interval around it, and the two are complementary: a biased point estimate with an honest interval and an unbiased one with a dishonest interval fail a decision-maker in different ways.

A parallel line uses digital trace data. Blumenstock et al. (2015) predicted wealth from mobile phone metadata in Rwanda, and Chi et al. (2022) drew on several data sources at once, among them satellite imagery, mobile network records, topography and aggregated connectivity, to release

wealth microestimates covering every low- and middle-income country at 2.4 km resolution. Their training data came from surveys in 56 countries, with accuracy checked in 18 more. Smythe and Blumenstock (2022) used high-resolution poverty maps of this kind for geographic targeting of social assistance, which is the application that makes the reliability of these estimates consequential rather than academic.

Across this literature, headline performance is reported as R^2 or rank correlation. Uncertainty is not ignored. Chi et al. (2022) publish a confidence interval for every microestimate, deriving it from a linear regression on absolute residuals, and are explicit that this regression has limited explanatory power ($R^2 = 0.09$ under spatially stratified cross-validation). Kakooei and Daoud (2024) go further in our direction, quantifying predictive uncertainty from deep learning and evaluating which source countries to train on when predicting a target country, which is the same transfer setting we study.

One study closes that loop, concurrently with ours. Olanrewaju and Adebayo (2026) build relative wealth estimates for 23 African countries from open data, evaluate cross-country transfer at $R^2 = 0.62$ under leave-one-country-out, and report that their split-conformal intervals attain 88.4% coverage against a 90% target. That coverage figure accompanies estimates fitted on each country’s own survey, and their transfer analysis reports point accuracy without a matching coverage figure.

A closer methodological neighbour is Adjei et al. (2026), who fuse satellite and reanalysis data to map PM2.5 across 29 African countries, validate under location-grouped spatial cross-validation, and apply split conformal prediction at a nominal 90% level. They report empirical coverage and find it fails badly under geographic shift, at 65.3% against the 90% target in East Africa, which they attribute to covariate shift measured directly. The outcome variable is air quality rather than welfare, the shift is regional rather than across a national border, and coverage is reported at one nominal level for one region. Their result nonetheless matters here, because it establishes that conformal coverage on African satellite products can collapse under spatial shift. Our finding that it largely holds is therefore a contrast to be explained rather than a foregone conclusion, and we return to it in Section 6.

What remains unmeasured is therefore not whether these intervals are calibrated at all, but whether calibration survives the border for this outcome. Establishing that requires coverage in-country and out-of-country for the same model and the same calibration set, resolved per country rather than pooled, and at more than one nominal level, since a method can hold at 90% while failing at 50%. We report all three, with the target country held back for a single pre-registered test. Where our figures and theirs are comparable they agree closely, which we read as independent corroboration rather than as a discrepancy to explain.

2.2 Small area estimation, where coverage is validated

There is a separate tradition in which interval calibration is standard. Model-based small area estimation combines survey data with auxiliary covariates and reports mean squared error and interval coverage as a matter of course, and it has increasingly used geospatial and satellite covariates for poverty and wealth (Masaki et al., 2020; Newhouse, 2023; Permatasari and Ubaidillah, 2025). Coverage in that literature is typically established by design-based simulation or by comparison against a census, in settings where the model is applied within the country whose survey trained it.

Our question is what happens to a distribution-free interval when the model is carried across a border into a country with no survey at all, which is the regime the satellite-based literature actually targets and the one small area estimation is not usually asked to serve. We take the practice of validating coverage from that tradition and apply it in this one.

2.3 Validation under spatial dependence

Survey clusters near one another share covariates, so a random train/test split places near neighbours on both sides and inflates measured accuracy. Roberts et al. (2017) survey blocking schemes for data carrying spatial, temporal, hierarchical or phylogenetic dependence. Their recommendation is stronger than it first appears: blocking should be used whenever the design implies dependence, and a model whose residuals look uncorrelated is not evidence that it can be skipped. Ploton et al. (2020) give a case where the correction is not marginal. Mapping forest biomass in central Africa, they report that the usual random-split figure credited the model with explaining over half the variation, and that once folds were separated in space almost none of that apparent skill remained.

The reason this bears on our design is specific. Our central comparison is between in-country and out-of-country performance, so an inflated in-country baseline would manufacture the effect we set out to measure. We therefore block all within-country validation and report the naive figure alongside, so the size of the correction is visible.

2.4 Conformal prediction and distribution shift

Conformal prediction (Vovk et al., 2005) constructs prediction sets with finite-sample marginal coverage for any underlying model, requiring only that calibration and test data be exchangeable. Romano et al. (2019) introduced conformalized quantile regression, which conformalises the output of quantile models so that interval width varies with the input rather than being constant across the covariate space.

The exchangeability requirement is exactly what cross-border transfer violates. Tibshirani et al. (2019) generalised conformal prediction to weighted exchangeability, showing that valid coverage can be recovered under covariate shift when the density ratio between source and target is known or estimable. Barber et al. (2023) analysed the non-exchangeable case more broadly and bounded the coverage gap in terms of the distribution shift.

That theory tells us coverage *can* fail under shift and how it might be repaired. It does not say how much coverage is actually lost when a poverty model crosses a West African border, which is an empirical quantity. Measuring it is the contribution here. We deliberately use unweighted conformal prediction rather than the weighted correction of Tibshirani et al. (2019), because the question is what happens to the guarantee as it is ordinarily applied, not whether a correction exists.

2.5 Domain adaptation

Methods that align source and target feature distributions are the standard response to covariate shift. We use CORAL (Sun and Saenko, 2016), which matches second-order statistics between domains by a linear transformation and requires target features but no target labels. Its relevance to this paper is that distribution alignment operates on features and has no mechanism acting on interval width, which motivated our third hypothesis.

3 Data

3.1 Surveys

Twelve West African DHS surveys, listed in Table 1. The unit of analysis is the DHS cluster, an enumeration area, which is the finest spatial unit at which wealth is released. The label is the cluster mean of hv271, the DHS wealth index factor score.

Country	Survey	Year used	Clusters	Households
The Gambia (target)	2019–20	2020	280	6,549
Senegal	2019 Continuous	2019	214	4,538
Mali	2018	2018	328	9,510
Nigeria	2018	2018	1,382	40,427
Ghana	2014	2014	423	11,835
Burkina Faso	2021	2021	514	13,251
Côte d’Ivoire	2021	2021	539	14,766
Mauritania	2019–21	2020	1,198	11,658
Sierra Leone	2019	2019	557	13,399
Guinea	2018	2018	401	7,912
Benin	2017–18	2017	540	14,156
Togo	2013–14	2014	330	9,549
Total			6,706	155,796

Table 1: Surveys used. The year is the modal interview year in hv007, which differs from the survey label for four surveys spanning more than one calendar year. Cluster counts exclude those DHS could not georeference.

Every survey is constrained to 2021 or earlier. Version 2.1 of the VIIRS annual nightlights composite covers 2012 to 2021 and version 2.2 covers 2022 onward; they are separate products rather than a reprocessing. Using the most recent survey for each country would have made nightlight product version correlate with country for four of the twelve, in the dimension this study measures. Constraining the sample keeps one product across all countries, at a cost in recency that is largest for Ghana (2022 to 2014).

3.2 The label is standardised within country

The wealth index is constructed by principal component analysis within each survey and is standardised to mean zero and unit standard deviation in each country. We verified this at household level: Nigeria 0.0000 and 0.9940, Sierra Leone 0.0000 and 0.9909, The Gambia 0.0000 and 0.9874. Across the twelve countries, the variance of cluster-mean wealth between countries is 0.0002 against a total of 0.7400.

This matters for interpretation. The label is a within-country relative position, so a cluster at +1 in Sierra Leone and one at +1 in Nigeria are not comparable in absolute terms. Transfer therefore requires a model to map absolute satellite measurements onto a country-relative rank whose scaling depends on the target country’s own distribution. This is a plausible mechanism for calibration failure and we return to it in Section 6.

3.3 Covariates and displacement

DHS displaces cluster coordinates before release, by up to 2 km in urban areas and 5 km in rural areas, with 1% of rural clusters displaced up to 10 km (Burgert et al., 2013). Extracting a covariate at the recorded point therefore reads the wrong location. We summarise each covariate over a buffer matched to the displacement distance, 2 km urban and 5 km rural, computed in a local UTM projection.

Rural buffers use 5 km rather than 10 km. The 1% displaced further are not identifiable in the released data, and a 10 km radius applied to every rural cluster would inflate 99% of them to

roughly four times the necessary area. We accept a known error on 1% of clusters in exchange for a sharper signal on the remainder.

Twelve features are used, all from open sources and extracted server-side in Google Earth Engine: VIIRS nightlights (mean, standard deviation, and the background-masked mean), WorldPop population (mean, standard deviation), Global Human Settlement built surface split into total and non-residential, ESA WorldCover land cover class (mode), CHIRPS annual rainfall, open water fraction, and SRTM elevation (mean, standard deviation). Survey metadata such as the urban/rural flag is excluded, since a model given it is partly reading the survey’s own classification of the place rather than the satellite record.

CHIRPS is a land-only product and returns no value over water. The Gambia is a narrow strip of land around a river and has an unusually high share of water-adjacent clusters, so unhandled masking would remove target-country clusters at a higher rate than training-country ones and introduce a distribution shift of our own making. We fill masked pixels with the mean of land rainfall within 20 km and carry water fraction as a covariate so the affected clusters can be identified.

4 Design

4.1 Pre-registration

The protocol, hypotheses, falsification criteria, splitting scheme and sanity gate bands were written and committed to version control before data access was granted. The specification for the single target-country evaluation, including its threshold and the wording required for each outcome, was committed at `bfa32ce` before any Gambian wealth label was read. Amendments are dated, state what changed and why, and leave the original text in place.

4.2 Splitting

DHS clusters are spatially autocorrelated, so a random train/test split places near neighbours on both sides and inflates measured accuracy (Roberts et al., 2017; Ploton et al., 2020). Since our central comparison is between in-country and out-of-country performance, an inflated in-country baseline would manufacture the result we were looking for. All within-country validation uses spatially blocked k -fold cross-validation, with whole blocks held out and block size set to twice the empirical variogram range (55 km, giving 110 km blocks).

Cross-country estimates use leave-one-country-out over the eleven training countries, which gives eleven independent country pairs rather than one. The Gambia appears in no training, calibration or model-selection step.

4.3 Uncertainty

Split conformal prediction with absolute-residual scores (Vovk et al., 2005) gives a finite-sample coverage guarantee under exchangeability between calibration and test data. Cross-border transfer is precisely where exchangeability fails, which is why we chose a method with a proof attached: watching a guarantee lose its coverage is easier to attribute than watching a heuristic degrade.

We also fit conformalized quantile regression (CQR, Romano et al., 2019), whose interval width varies with the input. This was included in advance because adaptivity is the property most likely to survive a distribution shift, so a divergence between the two methods would indicate a failure of interval shape rather than of level.

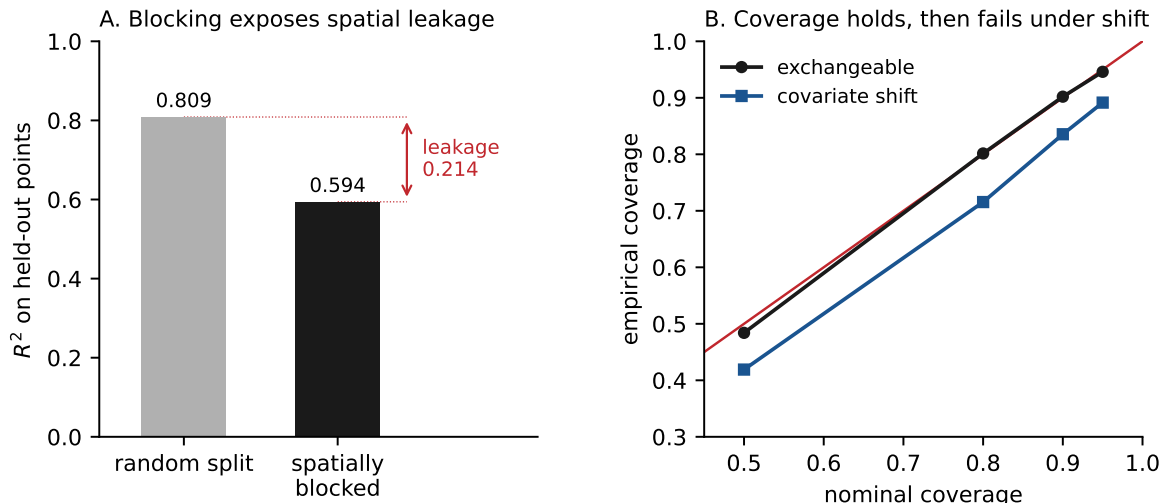


Figure 1: Simulation checks. **A**: points carry an unobserved short-range field absent from the covariates, so leakage exists by construction. A random split scores $R^2 = 0.809$ and a spatially blocked split scores 0.594, a gap of 0.214, which is the leakage the blocking is meant to expose. **B**: split conformal calibrated on one distribution and tested on the same one attains its nominal level, and loses coverage when the test distribution is shifted. At nominal 90% coverage is 0.902 under exchangeability and 0.836 under shift.

The calibration set is drawn only from training countries. Calibrating on the target would restore coverage by construction, and our implementation raises an error if target-country data reaches the calibration step.

4.4 Simulation checks on the two instruments

Both experiments rest on measurements whose correctness cannot be verified against the survey data, because there the truth is what we are trying to estimate. On synthetic data the truth is known, so each instrument was checked against a case where the right answer is fixed in advance. Neither check uses DHS data and both are deterministic given their seeds.

Does the blocking expose leakage? Points are placed on a jittered grid. The observed covariates are smooth, long-wavelength spatial fields, so a point’s covariate vector effectively encodes where it is. The target adds a short-wavelength field that is *not* among the covariates, standing in for local conditions no covariate set measures.

Under a random split a flexible model can locate a test point from its covariates, find its near neighbours in the training set, and recover their unobserved local conditions. Under a blocked split those neighbours are absent. Leakage is therefore present by construction, and a blocking scheme that fails to reveal it is not working. The gap is 0.214 (Figure 1A), which is the same order as the gaps we observe per country on the real data, 0.070 to 0.370.

Does coverage behave as the theory says? A model is fitted, then calibrated on data from one distribution and tested twice: once on data from that same distribution, and once on data whose covariates are shifted. Noise is heteroscedastic, so a constant-width interval is not automatically correct and the calibration set has to do real work.

Setting	R^2
In-country, random split (naive)	0.783
In-country, spatially blocked	0.746
Out-of-country, leave-one-country-out (mean of 11)	0.723
Target country, The Gambia	0.673

Table 2: Point accuracy. The gap between random and blocked in-country splits quantifies spatial leakage in the naive estimate.

Coverage attains its nominal level when calibration and test data are exchangeable, which is what the guarantee promises, and falls when they are not, which is what the guarantee permits (Figure 1B). This is the mechanism the paper goes on to look for across a national border, demonstrated where the shift is imposed rather than inferred.

Neither check validates the finding. They establish that the instruments respond as intended to effects of known sign and rough magnitude, so that a null result on the real data can be read as evidence about the data rather than as a silent failure of the measurement.

4.5 Sanity gate

Before any transfer claim we required the pipeline to reproduce an in-country accuracy figure consistent with published work, with bands fixed in advance. Pooled in-country blocked R^2 was 0.746, above the original ceiling of 0.70. That ceiling was a proxy for spatial leakage, adopted when no direct test was available. Three direct tests were then run and none supported leakage: R^2 falls only from 0.746 to 0.706 as block size increases sixfold from 110 km to 660 km; leave-one-country-out, which removes every spatial neighbour, gives 0.723; and no block was split across folds. The ceiling was also benchmarked against held-out-country CNN results, while the gate measures in-country blocked prediction, whose published comparator is $R^2 = 0.69$.

We amended the upper band after seeing 0.746, and the amendment says so. The band now passes results between 0.70 and 0.85 only when the three diagnostics are run and reported, and the implementation computes them rather than accepting an assertion. The original band and the original failure remain in the record.

We also validated the covariate extraction against the DHS Spatial Data Repository’s independently computed cluster covariates. Agreement was moderate (Spearman 0.727 to 0.963 on nightlights) and below our pre-registered threshold of 0.90. We report this as a failure. Two internal checks bound its interpretation: our WorldPop and GHSL summaries, which come from independent products through the same buffers and code, agree at Spearman 0.907 to 0.951 in all twelve countries, which is inconsistent with broken buffer geometry or a broken cluster join; and the DHS composite carries no year, while our extraction is matched to survey year.

5 Results

5.1 Point accuracy transfers

H1 predicted an out-of-country drop of roughly a third. The observed drop from in-country blocked to leave-one-country-out is 0.023, about three per cent (Table 2). H1 is falsified. This is consistent with Yeh et al. (2020), who already report that point accuracy survives a border.

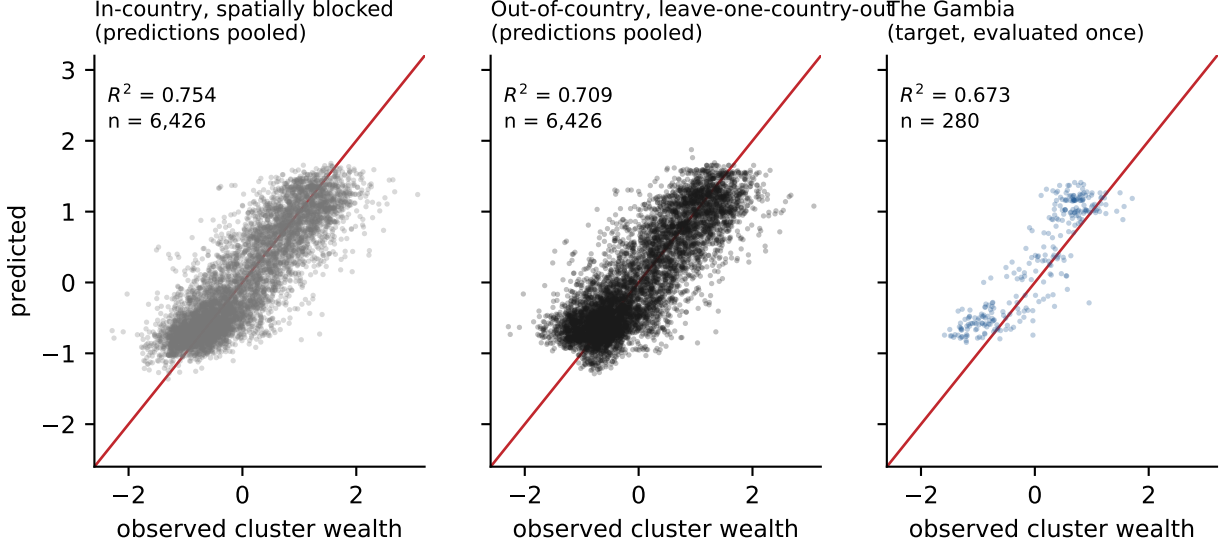


Figure 2: Predicted against observed cluster wealth, with the identity line in red. The target panel reproduces the registered evaluation exactly, fitted on folds 3 to 9 with folds 0 to 2 reserved for conformal calibration; fitting on all eleven training countries instead would give 0.696. The first two panels pool predictions across folds and countries, so their R^2 differs slightly from Table 2, which reports means across folds and across countries respectively. Predictions are visibly compressed toward the mean in all three panels, over-predicting the poorest clusters and under-predicting the richest, which is the pattern Corral et al. (2025) report for point estimates in unsurveyed areas.

5.2 Mean coverage transfers

H2, the central claim, predicted that a nominal 90% interval would cover substantially less than 90% out-of-country. It covered 88.7% across the eleven held-out training countries and 89.3% on The Gambia (Table 3). H2 is not supported.

The pre-registered target test, H4, asked whether Gambian coverage at nominal 90% would fall outside $[0.865, 0.935]$, the two-sided 95% binomial interval at $n = 280$ under exactly nominal coverage. It did not (0.893, $z = -0.40$). H4 is not confirmed. The test has power 0.529 against a false positive rate of 0.050, stated in advance, so this null is weak evidence and should not be read as evidence that coverage is reliable.

5.3 Correlation alignment does not help

H3 predicted that domain adaptation would improve point accuracy while leaving calibration un-repaired. CORAL (Sun and Saenko, 2016) degraded point accuracy, from $R^2 = 0.709$ to 0.646 out-of-country, and worsened the interval score for both conformal methods.

The baseline here is 0.709 rather than the 0.723 in Table 2 because the two are computed on different training sets. Table 2 fits each leave-one-country-out model on all ten remaining countries. The conformal experiments must reserve part of the source data to calibrate on, so they fit on five of ten spatial folds and hold three for calibration, which is roughly half the data and costs about 0.014 in R^2 . Both figures are reported rather than reconciled to one, since each is the correct baseline for its own comparison. The premise fails, so the comparison H3 proposed cannot be made. The reasoning behind H3 is nonetheless consistent with what we observe: aligning feature distributions

Nominal	Training pool (mean of 11)			The Gambia	
	In-country	Out-of-country	Shortfall	Split	CQR
0.50	0.524	0.491	0.033	0.371	0.475
0.80	0.825	0.785	0.041	0.704	0.775
0.90	0.913	0.887	0.026	0.893	0.896
0.95	0.956	0.943	0.013	0.961	0.968

Table 3: Empirical coverage of conformal prediction intervals. Training-pool columns use split conformal. Gambian columns are the single pre-registered evaluation.

contains no mechanism acting on interval width, and the dispersion of coverage across countries was unchanged (0.053 to 0.055 for split conformal, 0.060 to 0.062 for CQR). Only second-moment alignment was tested; the adversarial variant named in the protocol was not run.

5.4 Two post-hoc observations

The following were found after the pre-registered questions had been answered. They are reported as leads.

Coverage varies across countries more than sampling noise permits. At nominal 90%, out-of-country coverage ranged from 0.821 (Mauritania) to 0.979 (Togo), with standard deviation 0.053 against 0.019 in-country (Figures 5 and 6). If true coverage were exactly 0.90 everywhere, binomial sampling at these cluster counts would give 0.014, so observed dispersion is 3.7 times that ($\chi^2 = 195.1$, 10 d.f.). Nine of eleven countries fall outside an 88 to 92% band. The average therefore conceals substantial per-country variation, which matters because a practitioner applies a model to one country rather than to eleven. H4 was the out-of-sample test of this observation and returned a null, though at 53% power.

Constant-width intervals appear to fail in shape. On The Gambia, split conformal undercovered at nominal 50% (0.371, $z = -4.30$) and 80% (0.704, $z = -4.03$) while remaining indistinguishable from nominal at 90% and 95% (Figure 6, bottom row). CQR was within 1.4 standard errors at every level. One of six pre-specified secondary tests fell outside its band, which multiplicity alone roughly predicts, but the magnitude does not fit that explanation. Coverage that fails in the middle of the distribution and holds in the tails is characteristic of a width of the wrong shape rather than a width uniformly too small, which is the axis separating constant-width from adaptive intervals. The comparison was set up in advance for this reason, though without a registered threshold. The training pool showed no comparable low-level undercoverage on average (0.491 at nominal 0.50, 0.785 at nominal 0.80).

6 Discussion

Our pre-registered expectations were that transfer would be harmful and that uncertainty would be the first thing to break. Neither held. Across eleven country pairs and one held-out target, point accuracy degraded by about three per cent and mean interval coverage stayed within about one and a half points of nominal.

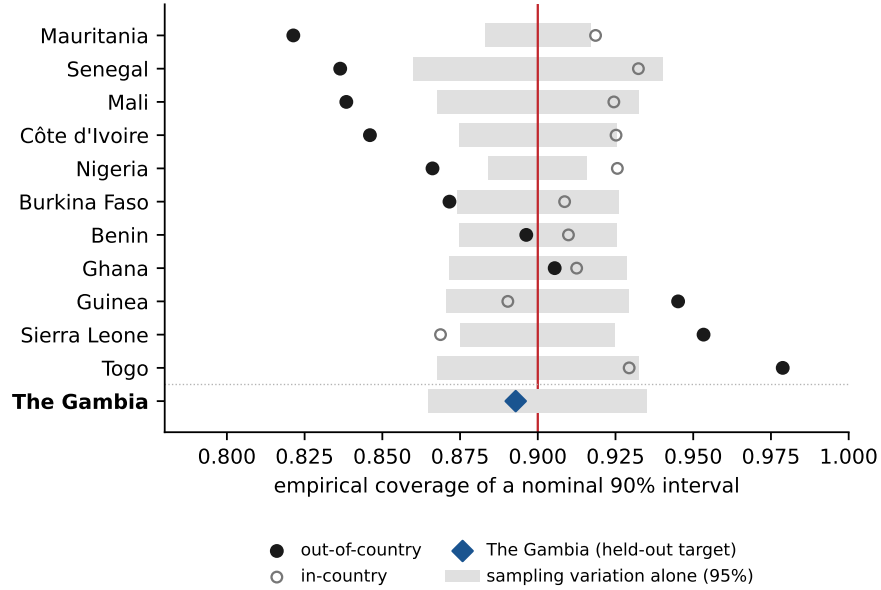


Figure 3: Empirical coverage of a nominal 90% split-conformal interval, by held-out country. Grey bars show the interval within which sampling variation alone would place the estimate if true coverage were exactly 0.90, given that country’s cluster count. Nine of eleven training countries fall outside a two point band around nominal, and observed dispersion is 3.7 times what sampling permits. The Gambia, evaluated once against a pre-registered threshold, is an unremarkable member of the spread.

We take the practical implication to be that these models transfer better than caution about distribution shift would suggest, at least within a region of neighbouring countries with a common survey instrument.

That conclusion should be read against Adjei et al. (2026), who measure conformal coverage on an African satellite product under spatial shift and find it falls to 65.3% against a 90% target. Coverage clearly can collapse in this setting, so the question is what differs here. Three things plausibly do. Their shift is between regions of a continent while ours is between adjacent countries sharing a survey instrument, so the covariate distributions are closer. Their label is a physical concentration with an absolute scale, while ours is standardised within each country by construction, which removes level differences from the label before any model sees it. And their in-domain accuracy is much lower ($R^2 = 0.134$ under spatial cross-validation against our 0.746), so their intervals are calibrated on residuals from a weaker model. We cannot separate these with the data here, and the comparison is a reason to treat our result as specific to this setting rather than as a general finding about conformal prediction under geographic shift. Two qualifications follow from the post-hoc observations. Average coverage across many countries is a poor summary of what a user of one country’s estimates should expect, since per-country coverage varied over roughly 0.82 to 0.98. And the choice between constant-width and adaptive conformal intervals may matter more under shift than in-domain, since the two methods agreed in-domain and diverged on the target at the lower nominal levels.

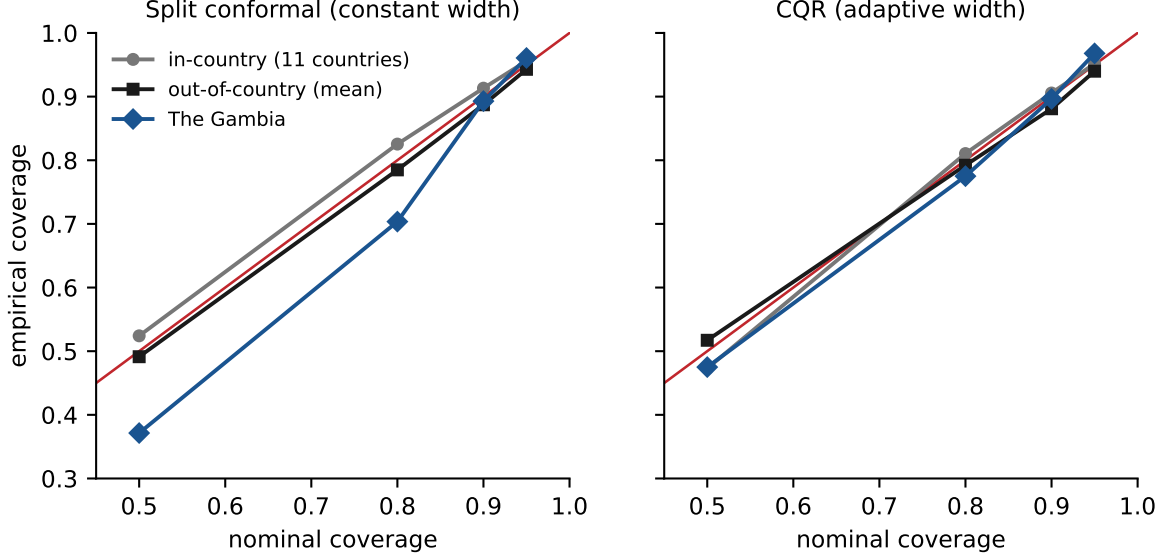


Figure 4: Calibration curves. The two conformal methods agree closely in-domain and out-of-domain on the training pool. On The Gambia they separate at the lower nominal levels: constant-width split conformal undercovers at nominal 50% and 80% while adaptive CQR does not. Coverage that fails in the middle of the distribution and holds in the tails is characteristic of a width of the wrong shape rather than one uniformly too small.

6.1 Limitations

Our source countries are neighbours of the target and share a survey instrument, so this measures transfer under a modest distribution shift. Transfer to a more distant region would plausibly be worse, making these results a lower bound on the problem.

The wealth index is an asset index standardised within country, not a measure of consumption or income, and its cross-country comparability is assumed rather than demonstrated. Mauritania sampled about ten households per cluster where other surveys sampled twenty to thirty, so its cluster means carry roughly 1.6 times the label noise, and this is correlated with country. Land cover comes from a single 2021 epoch applied to every survey year, which for Togo 2014 is an eight-year gap. The cross-check against DHS’s own covariate extraction did not reach our pre-registered agreement threshold.

The target-country test had 53% power. This is a consequence of having one target country with 280 clusters and was stated before the test was run. A study designed to test the dispersion observation would use many countries and every nominal level rather than a single draw.

6.2 A geographic pattern in the failures

Figure 5 suggests the coverage failures are not scattered at random. The three driest, most northerly countries undercover, Mauritania at 0.821, Senegal at 0.836 and Mali at 0.838, while three of the wetter coastal countries overcover, Togo at 0.979, Sierra Leone at 0.953 and Guinea at 0.945.

Tested after the fact, country mean latitude correlates with coverage at Spearman $\rho = -0.564$ ($p = 0.071$) and mean annual rainfall at $\rho = 0.655$ ($p = 0.029$). Two things make this weak evidence and both are stated rather than left for a reader to notice. The pattern was found by looking at the map, and the tests were run afterwards on the same eleven numbers that suggested it. Two

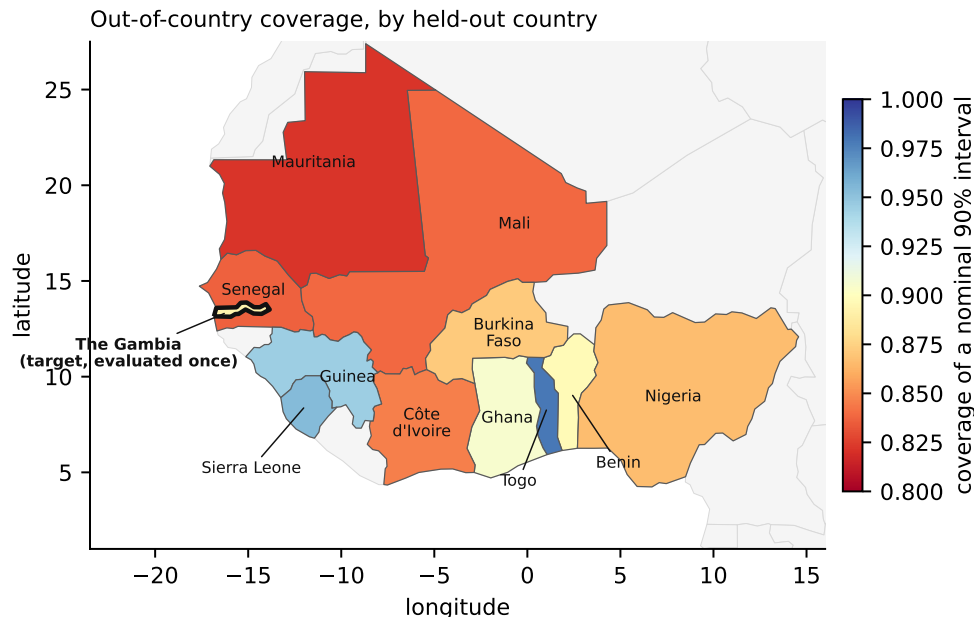


Figure 5: The twelve surveys, shaded by out-of-country coverage of a nominal 90% interval. The Gambia is outlined and was evaluated once. The failures are not uniformly distributed: the Sahelian countries at the top of the map undercover and the wetter coastal countries overcover. That apparent gradient was noticed by inspecting this figure and is treated as a lead in Section 6, not as a result.

covariates were tried, so the smaller p value should be read with that in mind, and the rank test on latitude does not reach conventional significance in any case.

We report it because it is a specific, testable proposition for a study designed around it, and because a reader looking at Figure 5 would see it anyway. It is not a finding of this paper.

6.3 Future work

The shape observation is the more tractable lead. It has a mechanism, it was anticipated in the design, and unlike the dispersion observation it can be tested with real power: constant-width against adaptive conformal, per country, at every nominal level, across all eleven training countries. That design has not been run and would require pre-registration on the same terms as H4.

7 Conclusion

We pre-registered four hypotheses about how satellite-based poverty models fail across a national border and found no support for any of them. Point accuracy and mean interval coverage both transferred, and correlation alignment made accuracy worse. Reporting this is more useful than reporting the paper we expected to write, because the calibration of these intervals had not previously been tested empirically, and for average coverage the answer appears to be that it largely holds.

What remains open is narrower and better specified than where we started: coverage reliability varies substantially between countries, and constant-width conformal intervals may misstate their shape under shift in a way adaptive intervals do not.

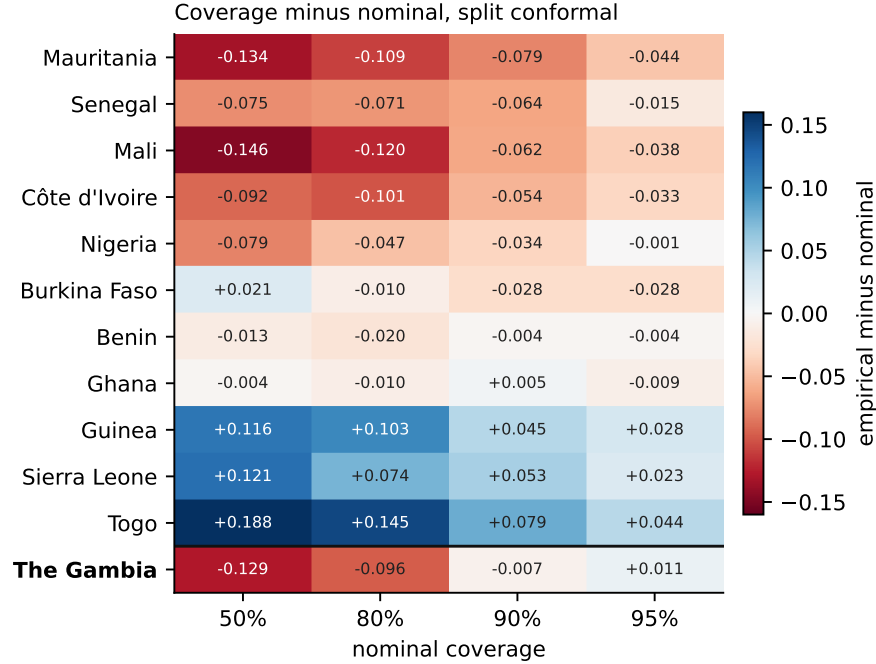


Figure 6: Empirical minus nominal coverage, split conformal, by country and nominal level. Both post-hoc observations are visible at once. Reading down a column gives the dispersion across countries, from Mali at -0.146 to Togo at $+0.188$ at nominal 50%. Reading along a row gives the level dependence: for almost every country the deviation shrinks as the nominal level rises. The Gambia, below the rule, shows that pattern clearly, at -0.129 , -0.096 , -0.007 and $+0.011$.

Data and code availability

All code is available at <https://github.com/Balisa50/gambia-poverty-transfer>, including the protocol, the dated amendments, the pre-registration and the evaluation script. DHS microdata cannot be redistributed under the terms of use and is not included; it is available on application at <https://dhsprogram.com>. All covariates are from open sources requiring no application.

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References

- Y. O. Adjei, D. Opoku, E. Abotsi, K. O. Amanqua, O. Kornyo, E. Soglo-Ahianyo, and C. A. Abbey. Conformal PM2.5 mapping under spatial covariate shift: satellite-reanalysis fusion for Africa's green industrial transition. arXiv:2604.22787, 2026. <https://arxiv.org/abs/2604.22787>
- R. F. Barber, E. J. Candès, A. Ramdas, and R. J. Tibshirani. Conformal prediction beyond exchangeability. *The Annals of Statistics*, 51(2):816–845, 2023. <https://doi.org/10.1214/23-AOS2276>

- J. Blumenstock, G. Cadamuro, and R. On. Predicting poverty and wealth from mobile phone metadata. *Science*, 350(6264):1073–1076, 2015. <https://doi.org/10.1126/science.aac4420>
- C. R. Burgert, J. Colston, T. Roy, and B. Zachary. Geographic displacement procedure and georeferenced data release policy for the Demographic and Health Surveys. DHS Spatial Analysis Reports No. 7. ICF International, Calverton, Maryland, 2013.
- G. Chi, H. Fang, S. Chatterjee, and J. E. Blumenstock. Microestimates of wealth for all low- and middle-income countries. *Proceedings of the National Academy of Sciences*, 119(3):e2113658119, 2022. <https://doi.org/10.1073/pnas.2113658119>
- M. Kakooei and A. Daoud. Increasing the confidence of predictive uncertainty: earth observations and deep learning for poverty estimation. *IEEE Transactions on Geoscience and Remote Sensing*, 62, 2024. <https://doi.org/10.1109/TGRS.2024.3392605>
- T. Masaki, D. Newhouse, A. R. Silwal, A. Bedada, and R. Engstrom. Small area estimation of non-monetary poverty with geospatial data. Policy Research Working Paper 9383. World Bank, Washington, DC, 2020. <https://doi.org/10.1596/1813-9450-9383>
- D. Newhouse. Small area estimation of poverty and wealth using geospatial data: what have we learned so far? Policy Research Working Paper 10512. World Bank, Washington, DC, 2023. <https://doi.org/10.1596/1813-9450-10512>
- A.-S. Olanrewaju and S. Adebayo. Measuring subnational welfare across sub-Saharan Africa: reproducible relative-wealth estimates from open satellite data. SSRN working paper 7086839, 2026. https://papers.ssrn.com/sol3/papers.cfm?abstract_id=7086839
- N. Permatasari and A. Ubaidillah. Small area estimation of poverty using remote sensing data. *Statistical Journal of the IAOS*, 41, 2025. <https://doi.org/10.1177/18747655241308390>
- P. Corral, H. Henderson, and S. Segovia. Poverty mapping in the age of machine learning. *Journal of Development Economics*, 172:103377, 2025. <https://doi.org/10.1016/j.jdeveco.2024.103377>
- K. Lee and J. Braithwaite. High-resolution poverty maps in sub-Saharan Africa. *World Development*, 159:106028, 2022. <https://doi.org/10.1016/j.worlddev.2022.106028>
- N. Jean, M. Burke, M. Xie, W. M. Davis, D. B. Lobell, and S. Ermon. Combining satellite imagery and machine learning to predict poverty. *Science*, 353(6301):790–794, 2016. <https://doi.org/10.1126/science.aaf7894>
- P. Ploton, F. Mortier, M. Réjou-Méchain, N. Barbier, N. Picard, V. Rossi, et al. Spatial validation reveals poor predictive performance of large-scale ecological mapping models. *Nature Communications*, 11(1):4540, 2020. <https://doi.org/10.1038/s41467-020-18321-y>
- D. R. Roberts, V. Bahn, S. Ciuti, M. S. Boyce, J. Elith, G. Guillera-Aroita, et al. Cross-validation strategies for data with temporal, spatial, hierarchical, or phylogenetic structure. *Ecography*, 40(8):913–929, 2017. <https://doi.org/10.1111/ecog.02881>
- Y. Romano, E. Patterson, and E. J. Candès. Conformalized quantile regression. In *Advances in Neural Information Processing Systems 32 (NeurIPS 2019)*, pages 3543–3553, 2019.
- I. S. Smythe and J. E. Blumenstock. Geographic microtargeting of social assistance with high-resolution poverty maps. *Proceedings of the National Academy of Sciences*, 119(32):e2120025119, 2022. <https://doi.org/10.1073/pnas.2120025119>
- B. Sun and K. Saenko. Deep CORAL: correlation alignment for deep domain adaptation. In *Computer Vision – ECCV 2016 Workshops*, volume 9915 of *Lecture Notes in Computer Science*, pages 443–450. Springer, 2016. https://doi.org/10.1007/978-3-319-49409-8_35
- R. J. Tibshirani, R. F. Barber, E. J. Candès, and A. Ramdas. Conformal prediction under covariate

- shift. In *Advances in Neural Information Processing Systems 32 (NeurIPS 2019)*, pages 2530–2540, 2019.
- V. Vovk, A. Gammerman, and G. Shafer. *Algorithmic Learning in a Random World*. Springer, New York, 2005.
- C. Yeh, A. Perez, A. Driscoll, G. Azzari, Z. Tang, D. Lobell, S. Ermon, and M. Burke. Using publicly available satellite imagery and deep learning to understand economic well-being in Africa. *Nature Communications*, 11(1):2583, 2020. <https://doi.org/10.1038/s41467-020-16185-w>